

Finite-Time Feedback Control through a Blackboard

Sampled message exposure, susceptibility, availability normalization, information transmission, and path irreversibility

8 September 2026

Abstract

We reformulate the earlier finite-time single-affinity feedback report for the blackboard interaction protocol. The observable of interest remains the intervention-versus-silence susceptibility, but the actuation model changes in one essential way: posting b controller messages does not create b guaranteed controlled updates. Instead, each of the M population updates samples q messages from a board containing B eligible peer messages and b controller messages, so the intervention acts only through the probability that a focal agent is actually exposed to controller content. Both the intervention and silence branches therefore contain autonomous population dynamics.

A mixture of two finite-compliance kernels remains exactly solvable. This gives a closed-form blackboard susceptibility $\chi_b(x)$ and an availability-normalized companion $\chi_{b,\text{avail}}(x) = \chi_b(x)/(1-x)$ for $x < 1$, exact finite-state kernels Q_0 and Q_1 , the exact state-local action-to-population information $T_\pi(n) = I(U; n' | n)$, a bounded action-information fraction η_{IF} , and the previously used bounded information-response efficiency η_{IR} . We also derive a branch-affinity path identity

$$\Delta S_{\text{sys}} + I(n; Y) + \mathbb{E}[h_U(n' - n)] = \Sigma \geq 0,$$

which is exact for the coarse blackboard reference. Unlike the isolated controlled layer, however, the silent branch has its own dynamics and generally its own affinity. Therefore the non-storage path term contains baseline social transport as well as intervention-induced transport. A bounded total path-transport fraction can be defined, but it should *not* be called a controller thermodynamic efficiency in the general blackboard model. A controller-associated thermodynamic efficiency is recovered under an additional neutral-baseline condition $h_0 = 0$.

The conceptual chain used throughout is

$$\text{sensing/policy} \longrightarrow w_b \longrightarrow (\chi_b, \chi_{b,\text{avail}}) \longrightarrow T_\pi \longrightarrow (\eta_{\text{IF}}, \eta_{\text{IR}}) \longrightarrow J_c^{\text{ex}} \longrightarrow \Sigma.$$

The report is a solvable reference theory, not a fitted account of the current LLM experiments. The numerical figures use the illustrative blackboard parameters supplied in the technical note unless stated otherwise.

1 Problem and one-cycle architecture

Consider a population of N agents and a designated controller target Z . Let

$$n_k \in \{0, 1, \dots, N\}, \quad x_k = \frac{n_k}{N},$$

where n_k is the number of agents supporting Z immediately before sensing at feedback cycle k .

The causal controller loop is still

$$n_k \longrightarrow Y_k \xrightarrow{\pi} U_k \longrightarrow n_{k+1},$$

with $U_k \in \{0, 1\}$ denoting silence or active controller posting. The important blackboard distinction is inside the final arrow. When $U_k = 1$, the controller posts b messages, but a focal agent only sees controller content if one of those messages is included in its sampled blackboard context.

A round contains M asynchronous updates. Each update samples one focal agent and then samples q messages from the eligible board. We use the following variables and parameters:

Symbol	Meaning
N	population size
$n, x = n/N$	target-support count and share before the round
q_c	number of agents sampled by the controller sensor
Y	number of target supporters in the sensor sample
U	controller action: 0 = silence, 1 = post controller messages
M	number of population updates in one round, usually $M = N$
B	number of eligible peer messages on the frozen board
b	number of controller messages actually posted
q	number of board messages sampled by a focal update
w_b	probability that an update samples at least one controller message
(γ_0, p_0)	finite-compliance parameters of an unexposed/baseline update
(γ_1, p_1)	finite-compliance parameters of an exposed update
(γ_b, p_b)	effective parameters after marginalizing exposure
$\chi_b(x)$	intervention-minus-silence mean response in target-share units
$\chi_{b,\text{avail}}(x)$	susceptibility normalized by the remaining non-target share $1 - x$; diagnostic, not a bounded efficiency
$T_\pi(n)$	state-local action-to-population information
$\eta_{\text{IF}}(n)$	fraction of conditional action entropy visible in the next state
$\eta_{\text{IR}}(n)$	bounded information-response efficiency
J_c^{ex}	controller-excess target-count current relative to silence
Σ	finite-time KL irreversibility of the coarse feedback path

Basic modeling assumptions. The closed form requires a frozen board within a round, homogeneous effective response parameters, a binary distinction between “no controller message sampled” and “at least one controller message sampled”, and a shared unexposed baseline in the intervention and silence branches. These are deliberately strong reference assumptions. They are useful because they make falsifiable predictions; they are not meant to hide the richer LLM state.

2 Population ensembles and finite-time notation

Across repeated realizations, let

$$p_k(n) = \mathbb{P}(n_k = n), \quad p_{k+1}(m) = \mathbb{P}(n_{k+1} = m).$$

No stationarity is assumed. In general $p_{k+1} \neq p_k$.

For a local one-cycle derivation we suppress the cycle index and write

$$p(n) \equiv p_k(n), \quad p'(m) \equiv p_{k+1}(m).$$

For the illustrative finite-time current and path plots we use

$$p_{\text{init}}(n) = \text{Binomial}(N, 0.25),$$

only as a convenient ensemble. None of the exact state-local formulas depends on that choice.

The main numerical baseline used in the figures is

$$N = M = 24, \quad B = 18, \quad q = 3, \quad q_c = 12, \quad \beta = 4, \quad \theta = \frac{1}{2},$$

$$(\gamma_0, p_0) = (0.35, 0.70), \quad (\gamma_1, p_1) = (0.65, 0.90).$$

These are illustrative reference values taken from the supplied blackboard note, not fitted parameters for the current experiments.

3 Sensing and feedback decision

The controller samples q_c agents without replacement. If Y is the number of sampled agents supporting the target, then

$$Y \mid n \sim \text{Hypergeom}(N, n, q_c),$$

with

$$S(y \mid n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}.$$

The same soft feedback policy as in the original report is retained:

$$\pi(1 \mid y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

After averaging over the finite sensor,

$$a_n \equiv \mathbb{P}(U = 1 \mid n) = \sum_y S(y \mid n) \pi(1 \mid y).$$

The sensor and policy therefore determine whether the controller posts; the blackboard exposure model below determines whether a focal update actually sees those posts.

4 Blackboard exposure and finite-compliance microscopic response

4.1 Sampled exposure probability

Suppose an active controller posts b messages to a board with B eligible peer messages. A focal update samples

$$m = \min(q, B + b)$$

messages uniformly without replacement. The probability of at least one controller message is

$$w_b = 1 - \frac{\binom{B}{m}}{\binom{B+b}{m}}$$

with the convention $\binom{B}{m} = 0$ for $m > B$. For the common regime $q \leq B$, simply set $m = q$.

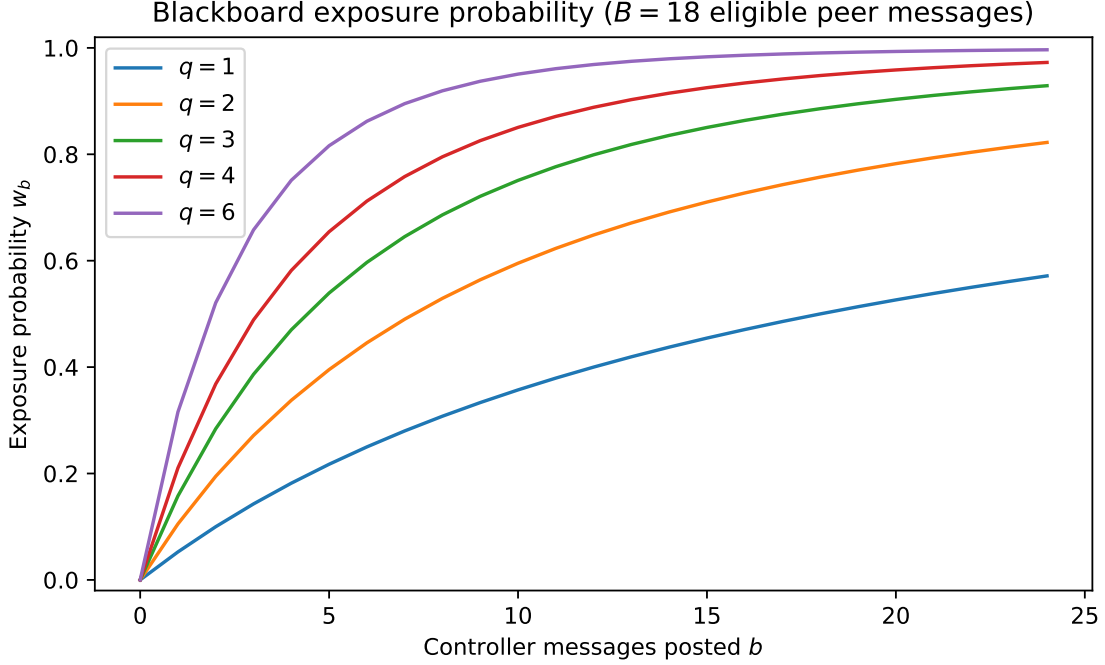


Figure 1: Exact exposure probability as a function of the number of controller messages. Message budget has diminishing returns because w_b saturates toward one. Crucially, b is a message count, not a count of guaranteed controlled updates.

For the technical-note example $B = 18$, $b = 9$, and $q = 3$,

$$w_9 = 0.7210256410.$$

Thus nine controller posts do not imply nine controlled revisions; they imply that each of the M updates is exposed with probability about 0.72 under the frozen-board approximation.

4.2 Two microscopic response classes

Use the finite-compliance kernel

$$K_{\gamma,p}(n, n+1) = \gamma \left(1 - \frac{n}{N}\right) p,$$

$$K_{\gamma,p}(n, n-1) = \gamma \frac{n}{N} (1-p),$$

with the diagonal chosen so rows sum to one. Here $\gamma \in [0, 1]$ is the probability of a revision opportunity and p is the effective preference conditional on revision. A revision may leave the vote unchanged.

Let

$$K_0 = K_{\gamma_0, p_0}$$

describe updates with no controller message in view, and

$$K_1 = K_{\gamma_1, p_1}$$

describe updates exposed to controller content.

4.3 Exact closure after marginalizing exposure

During an active round, an update is unexposed with probability $1 - w_b$ and exposed with probability w_b . Therefore

$$K_b = (1 - w_b)K_0 + w_bK_1.$$

Because the birth and death probabilities are linear in γ and γp , this mixture remains in the same family:

$$K_b = K_{\gamma_b, p_b}$$

with

$$\gamma_b = (1 - w_b)\gamma_0 + w_b\gamma_1, \quad p_b = \frac{(1 - w_b)\gamma_0 p_0 + w_b\gamma_1 p_1}{\gamma_b}.$$

When $\gamma_b = 0$, the kernel is the identity and p_b is immaterial.

The complete round kernels are now

$$Q_0 = K_0^M, \quad Q_1 = K_b^M.$$

This is the key structural change from the earlier report: *silence is no longer the identity operation*. Both branches evolve for M updates.

For $b = 9$ in the numerical baseline,

$$\gamma_b = 0.5663076923, \quad p_b = 0.8655166169.$$

5 Exact susceptibility: derivation of $\chi_b(x)$

5.1 Mean evolution under one finite-compliance kernel

For $K_{\gamma, p}$, if the current target share is x_j , then

$$\mathbb{E}[\Delta n \mid x_j] = \gamma(p - x_j),$$

so

$$\mathbb{E}[x_{j+1} \mid x_j] = \left(1 - \frac{\gamma}{N}\right) x_j + \frac{\gamma p}{N}.$$

Iterating M updates gives

$$F_M(x; \gamma, p) = p + (x - p) \left(1 - \frac{\gamma}{N}\right)^M.$$

This is the expected final target share after one round initialized at x .

5.2 Intervention minus silence

The blackboard susceptibility is defined exactly as before—the difference between the active and silent branches initialized from the same state:

$$\chi_b(x) = F_M(x; \gamma_b, p_b) - F_M(x; \gamma_0, p_0).$$

Writing

$$r_b = \left(1 - \frac{\gamma_b}{N}\right)^M, \quad r_0 = \left(1 - \frac{\gamma_0}{N}\right)^M,$$

we obtain the affine form

$$\chi_b(x) = \underbrace{p_b(1 - r_b) - p_0(1 - r_0)}_{c_b} + \underbrace{(r_b - r_0)}_{s_b} x.$$

Thus the model makes a strong prediction: at fixed effective parameters, blackboard susceptibility is affine in the initial target fraction. Arbitrary state dependence is not allowed by the reference.

If $r_b \neq r_0$, the zero-response point is

$$x_b^* = \frac{p_0(1 - r_0) - p_b(1 - r_b)}{r_b - r_0}.$$

It may lie outside $[0, 1]$, in which case the response has the same sign over all accessible states.

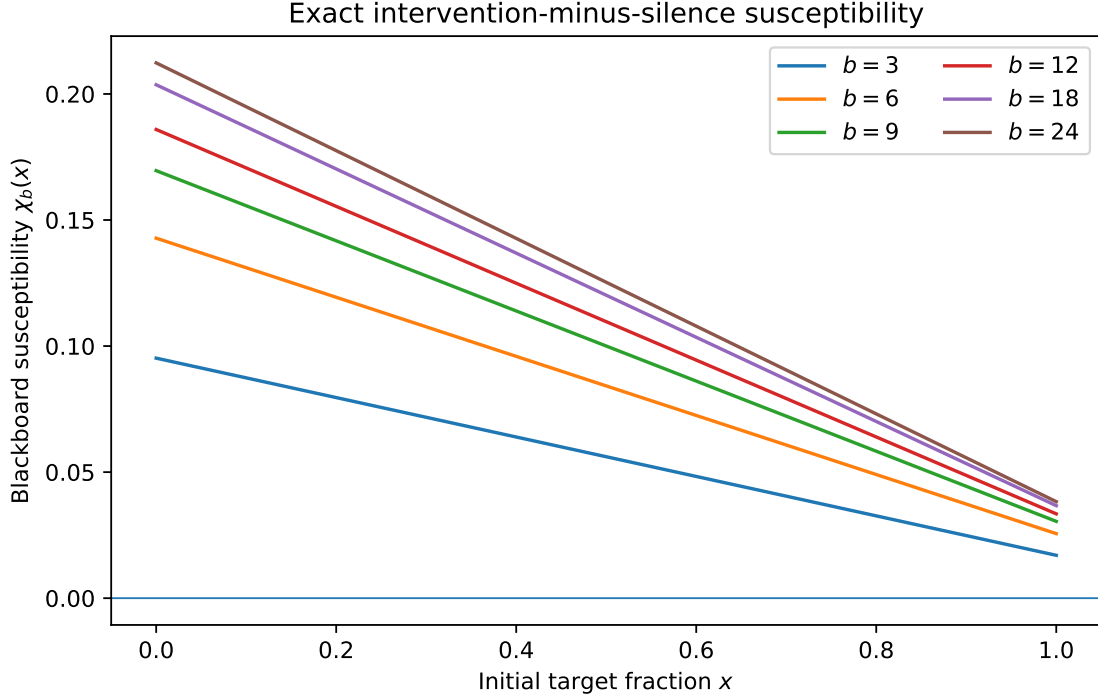


Figure 2: Exact blackboard susceptibility for several message budgets. Under the illustrative unequal-compliance parameters, the intervention has a larger effect at lower initial target support and saturates with message budget through the exposure probability.

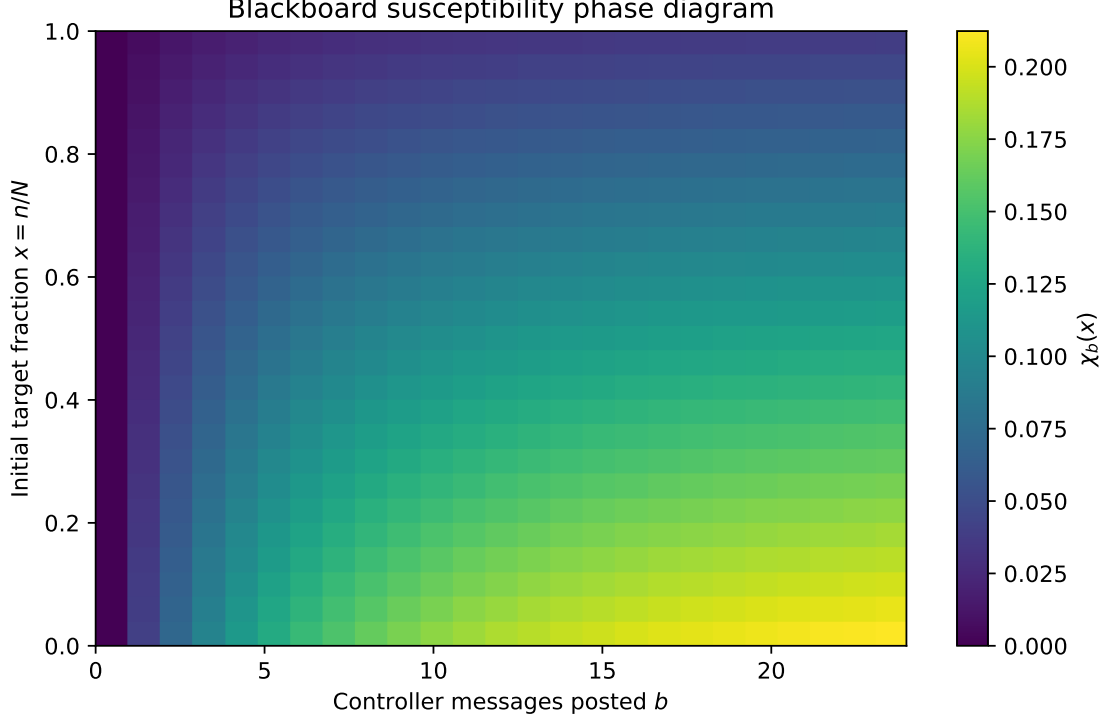


Figure 3: Susceptibility phase diagram over initial target share and message budget. The dominant blackboard signature is saturation in b rather than the direct “ b controlled opportunities” scaling of the isolated-actuation model.

5.3 Availability-normalized susceptibility

The raw susceptibility $\chi_b(x)$ is an absolute intervention-minus-silence change in target-share units. When the target already has large support, however, the remaining non-target population $1 - x$ is small. A useful companion diagnostic is therefore obtained by expressing the same branch contrast relative to this remaining population mass:

$$\chi_{b,\text{avail}}(x) = \frac{\chi_b(x)}{1 - x}, \quad x < 1.$$

The fully saturated state $x = 1$ is excluded because the denominator vanishes. This normalization answers a simple state-local question: *how large is the intervention-versus-silence response compared with the amount of non-target population that remains available?* It is particularly useful when comparing states with very different target support.

Substituting the exact blackboard susceptibility gives

$$\chi_{b,\text{avail}}(x) = \frac{p_b + (x - p_b) \left(1 - \frac{\gamma_b}{N}\right)^M - p_0 - (x - p_0) \left(1 - \frac{\gamma_0}{N}\right)^M}{1 - x}.$$

Equivalently, with $\chi_b(x) = c_b + s_b x$,

$$\chi_{b,\text{avail}}(x) = \frac{c_b + s_b x}{1 - x}.$$

Thus even an affine raw response can develop a strong high- x structure after availability normalization.

Interpretation caveat. $\chi_{b,\text{avail}}$ is *not* an efficiency and it is not constrained to $[0, 1]$. The numerator is a full-round difference between two evolving branches. It can contain both additional target adoption and differential retention of agents who already support the target. Consequently values above one are mathematically possible near saturation and should be read as a large branch contrast relative to a small remaining non-target mass, not as “more than 100% of available agents converted.” For this reason the raw susceptibility remains the primary response observable, while $\chi_{b,\text{avail}}$ is reported as a complementary normalization.

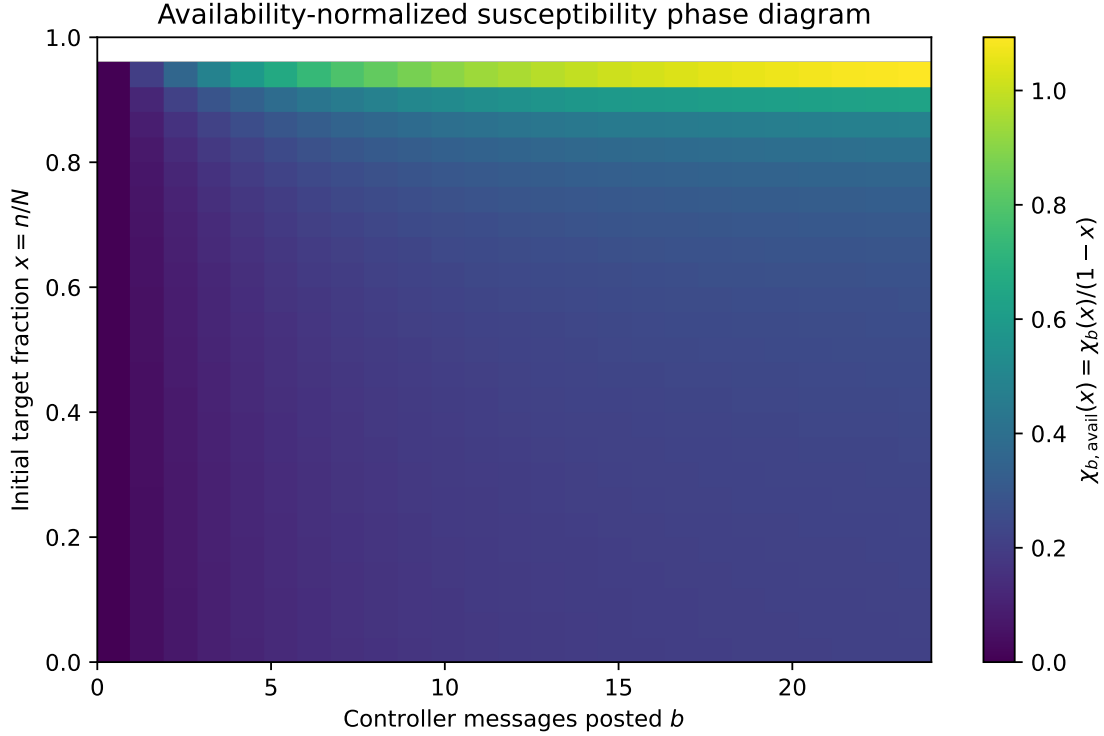


Figure 4: Availability-normalized susceptibility phase diagram. The state $x = 1$ is masked. Relative to the raw susceptibility heatmap, the high-target-support region is amplified because the same absolute branch contrast is compared with a smaller remaining non-target share $1 - x$.

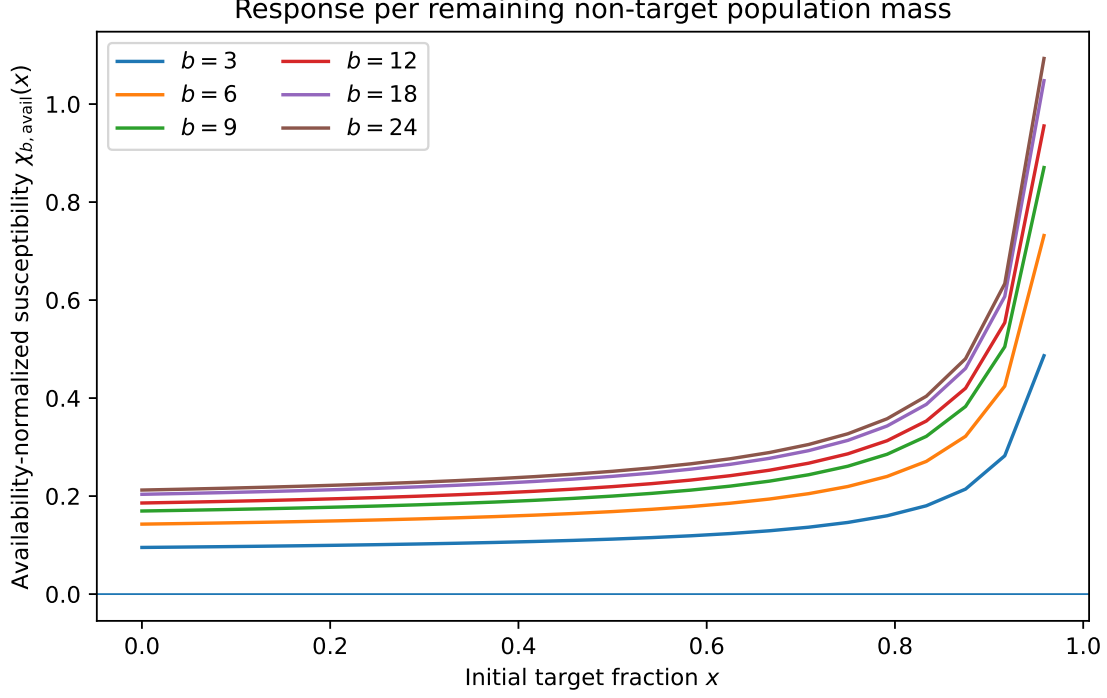


Figure 5: State-local availability-normalized susceptibility for representative message budgets. Increasing b raises exposure and hence the normalized response, while the $1/(1-x)$ normalization makes the response increasingly prominent as the remaining non-target population shrinks.

For compact summaries over an occupancy distribution $p(n)$, direct averaging of statewise ratios can overemphasize nearly saturated states. We therefore use the available-mass-weighted aggregate

$$\bar{\chi}_{\text{avail}}[p] = \frac{\sum_{n=0}^{N-1} p(n) \chi_b(n/N)}{\sum_{n=0}^{N-1} p(n) (1 - n/N)}.$$

This is the total occupancy-weighted branch response divided by the occupancy-weighted population mass that remains non-target, with $n = N$ excluded consistently from both sums.

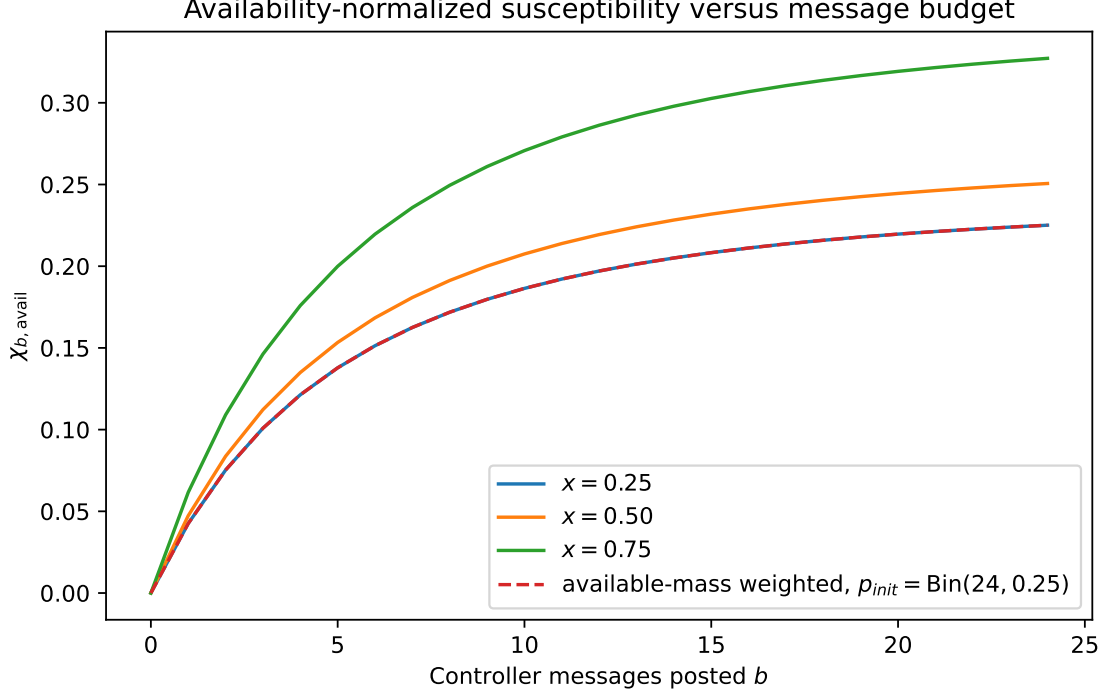


Figure 6: Availability-normalized susceptibility versus message budget at representative initial target shares, together with the available-mass-weighted aggregate under the illustrative Binomial(24, 0.25) initial ensemble. The curves retain the characteristic saturation in message budget inherited from the exposure probability w_b .

For the numerical baseline used throughout the report, representative values are:

b	w_b	$\chi_{b,\text{avail}}(0.25)$	$\chi_{b,\text{avail}}(0.50)$	$\chi_{b,\text{avail}}(0.75)$	$\bar{\chi}_{\text{avail}}[p_{\text{init}}]$
3	0.386	0.101	0.112	0.146	0.101
6	0.597	0.151	0.168	0.220	0.151
9	0.721	0.180	0.200	0.261	0.180
12	0.799	0.197	0.219	0.286	0.197
18	0.886	0.216	0.240	0.314	0.216
24	0.929	0.225	0.251	0.327	0.225

The table shows the two effects separately. Increasing b raises the response by increasing the chance of exposure, while at fixed b the same absolute response represents a larger perturbation relative to the remaining non-target mass when x is large. At the most extreme finite state $x = 23/24$, the baseline normalization reaches approximately 1.09 for $b = 24$; this is precisely why $\chi_{b,\text{avail}}$ must not be interpreted as a probability or bounded efficiency.

5.4 Equal-compliance special case

If $\gamma_0 = \gamma_1 = \gamma$, then $\gamma_b = \gamma$ and

$$p_b = (1 - w_b)p_0 + w_b p_1.$$

The x dependence cancels exactly:

$$\chi_b(x) = w_b(p_1 - p_0) \left[1 - \left(1 - \frac{\gamma}{N} \right)^M \right].$$

This is diagnostically valuable. If empirical susceptibility varies strongly with x while equal compliance is otherwise plausible, then at least one of the following must be true: exposed and unexposed revisions have different effective compliance, the hidden epistemic state changes with x , the board environment is not frozen, or the two-state coarse graining is insufficient.

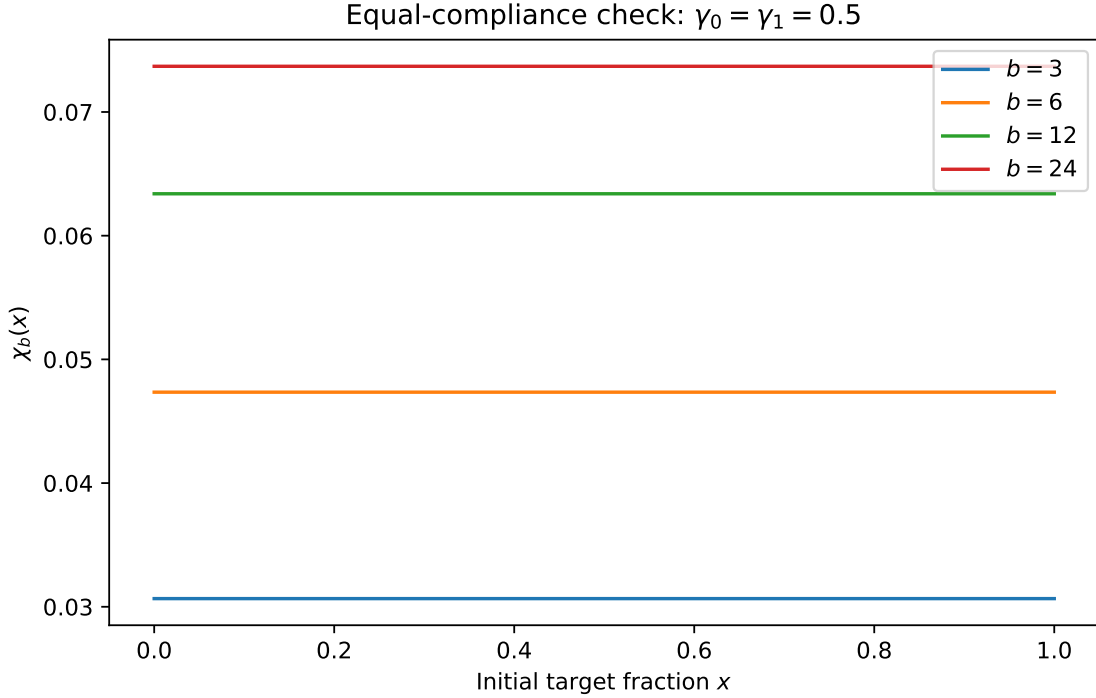


Figure 7: Equal-compliance structural check. Each budget produces a horizontal susceptibility curve. State dependence therefore becomes a direct model diagnostic rather than a free fitting degree of freedom.

5.5 A response-sign phase diagram

The two parameters of the exposed class play different roles. Increasing p_1 makes exposure more target-favoring, while increasing γ_1 makes the exposed update revise more readily. Their interaction can reverse the sign of the intervention effect when the exposed preference is weaker than the baseline preference or when kinetic acceleration moves the population more quickly toward an unfavorable set point.

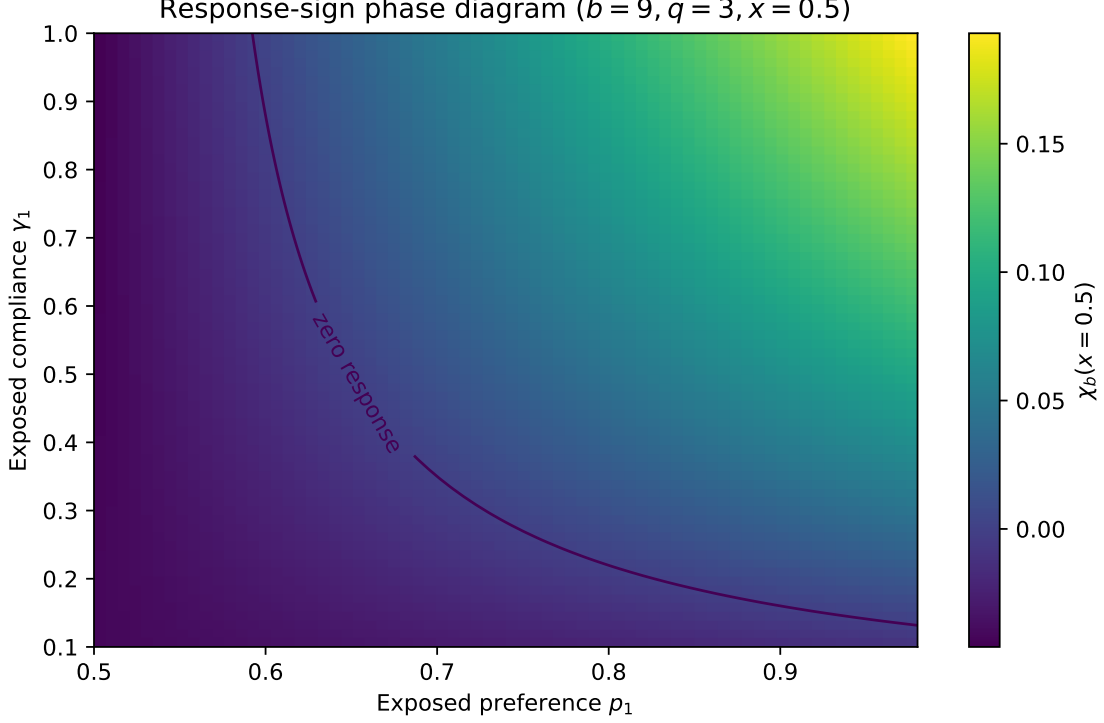


Figure 8: Response-sign phase diagram at $b = 9$, $q = 3$, and $x = 0.5$. The zero-response contour separates interventions that increase target support from interventions that reduce it. This makes explicit that larger exposure or faster revision does not by itself guarantee positive steering.

6 Exact state-local action-to-population information

6.1 Definition

At a fixed pre-round state n , the action-conditioned next-state kernels are

$$Q_0(m | n) = K_0^M(m | n), \quad Q_1(m | n) = K_b^M(m | n).$$

With $a_n = \mathbb{P}(U = 1 | n)$, the action-marginalized kernel is

$$Q_\pi(m | n) = (1 - a_n)Q_0(m | n) + a_nQ_1(m | n).$$

The exact state-local action-to-population information is

$$T_\pi(n) = I(U; n' | n) = \sum_{u \in \{0,1\}} \mathbb{P}(u | n) \sum_m Q_u(m | n) \log_2 \frac{Q_u(m | n)}{Q_\pi(m | n)}.$$

This is a weighted Jensen–Shannon divergence between the silent and active round kernels.

The interpretation is unchanged but the channel is now more realistic: T_π is large when (i) both controller actions occur with nontrivial probability and (ii) the full blackboard-evolved next-state distributions are distinguishable.

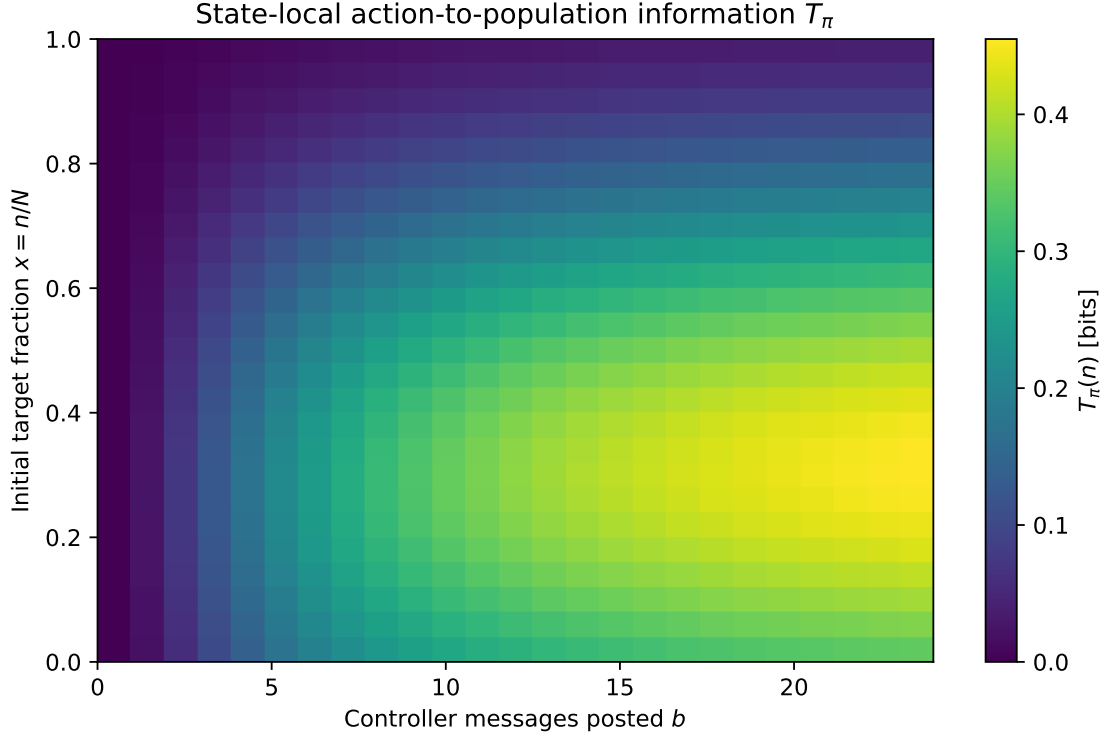


Figure 9: Exact state-local action-to-population information. The informative region is jointly shaped by blackboard exposure, baseline dynamics, and the state-dependent feedback policy.

6.2 Action-information fraction η_{IF}

Mutual information is bounded by the conditional entropy of the action:

$$0 \leq T_\pi(n) \leq H_2(a_n),$$

where

$$H_2(a) = -a \log_2 a - (1 - a) \log_2 (1 - a).$$

This gives the bounded information-fraction diagnostic

$$\eta_{\text{IF}}(n) = \frac{T_\pi(n)}{H_2(a_n)} \leq 1,$$

whenever $H_2(a_n) > 0$.

This quantity answers a narrower question than η_{IR} : *what fraction of the action uncertainty that remains at state n becomes recoverable from the next population state?* It measures action visibility, not coherent directional motion.

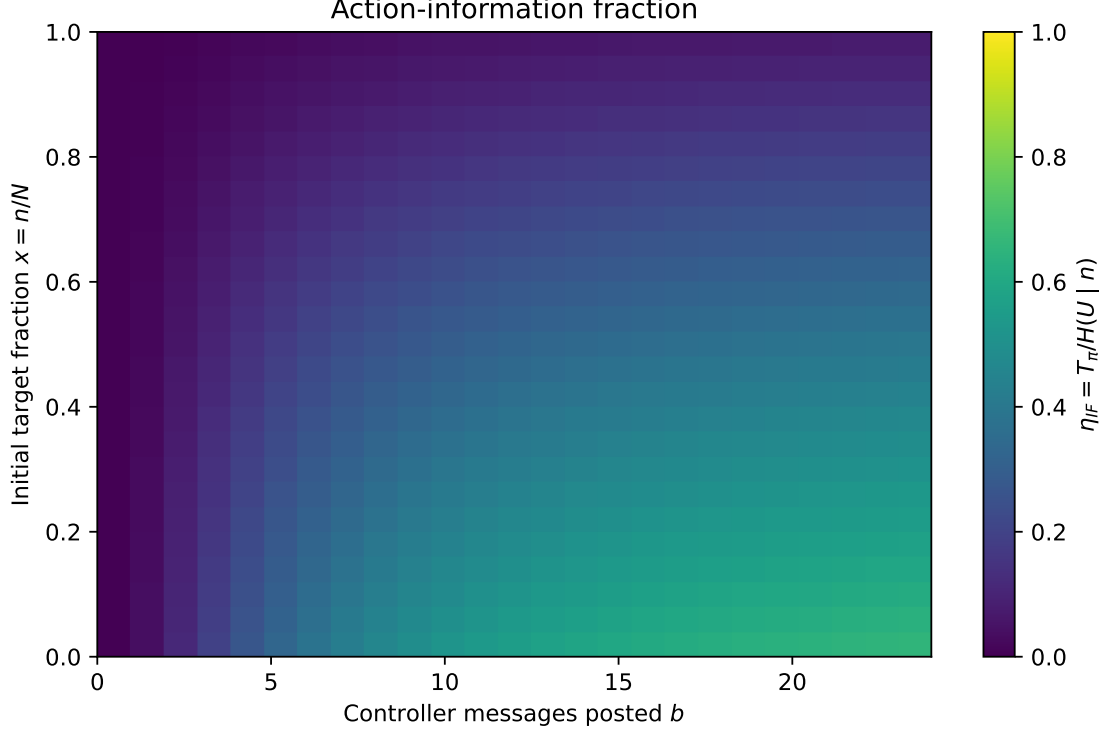


Figure 10: Action-information fraction. High values mean that a large fraction of the available conditional action entropy is visible in the next population state. The ratio can be high even when the mean susceptibility is not especially large.

6.3 Bounded information–response conversion

The Pinsker argument from the original report does not depend on Q_0 being the identity. At fixed n , let

$$\chi_b(n/N) = \mathbb{E}[x' \mid U = 1, n] - \mathbb{E}[x' \mid U = 0, n].$$

Because $x' = m/N \in [0, 1]$,

$$T_\pi(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_b(n/N)^2.$$

Hence

$$\eta_{\text{IR}}(n) = \frac{2a_n(1-a_n)\chi_b(n/N)^2}{(\ln 2)T_\pi(n)} \leq 1.$$

This remains a rigorous blackboard quantity. It asks how efficiently action-dependent distributional change is expressed as coherent mean target motion.

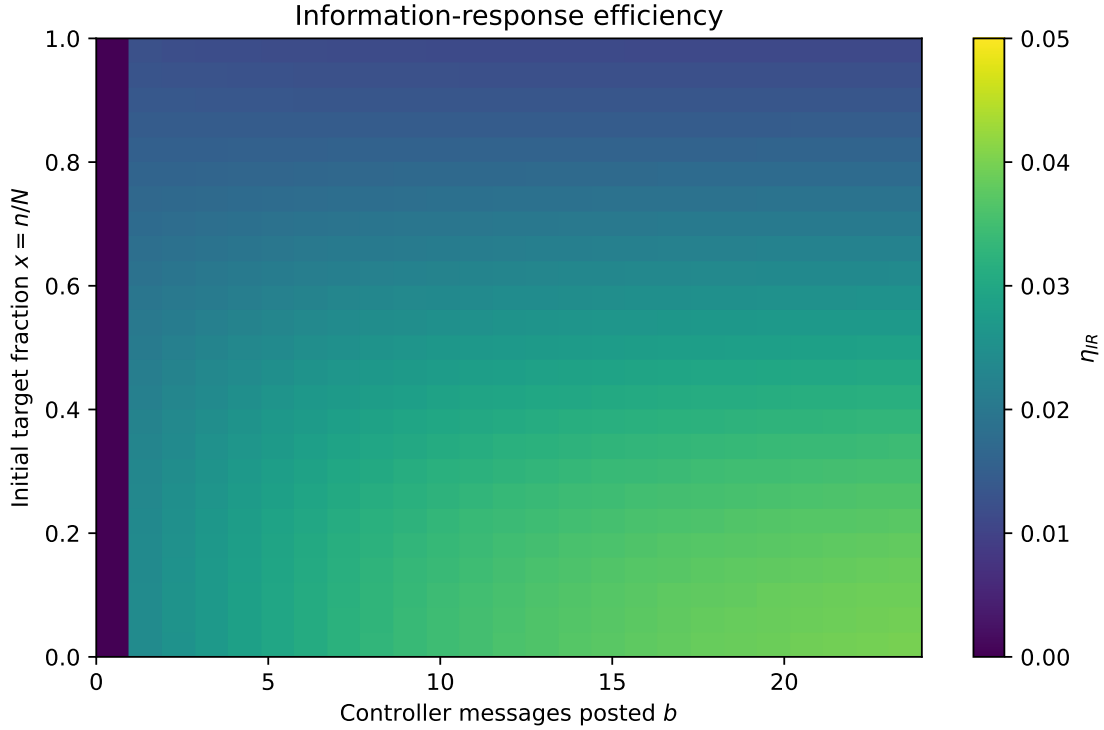


Figure 11: Information-response efficiency. In the illustrative baseline, η_{IR} is only a few percent: the action changes the distribution much more strongly than it changes its mean. This is precisely the type of distinction that T_π alone cannot reveal.

At $b = 9$, for example, the illustrative model gives approximately

x	$\chi_b(x)$	$T_\pi(x)$ [bits]	$(\eta_{\text{IF}}, \eta_{\text{IR}})$
0.25	0.135	0.328	(0.387, 0.032)
0.50	0.100	0.277	(0.277, 0.026)
0.75	0.065	0.134	(0.158, 0.018)

which makes the roles of the two bounded ratios concrete: action visibility can be substantial while the fraction expressed as coherent mean motion remains small.

6.4 Parameter sweeps

The blackboard model introduces two new resource axes before one even changes the response kernel: message budget b and sampled-message count q . Increasing either raises exposure, but with saturation. The sensor resource q_c remains distinct: it changes a_n rather than Q_0 or Q_1 directly.

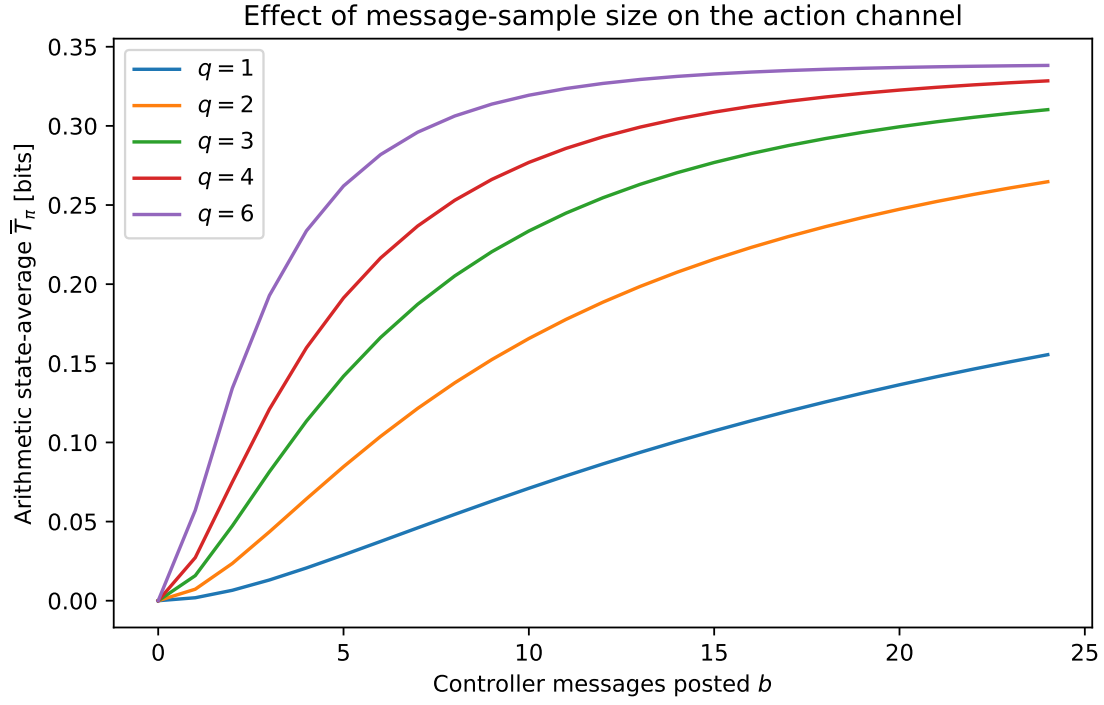


Figure 12: Arithmetic state-average $\bar{T}_\pi = (N + 1)^{-1} \sum_n T_\pi(n)$ while varying the number q of messages read by each focal update. Increasing q raises exposure and therefore strengthens the action channel, again with diminishing returns.

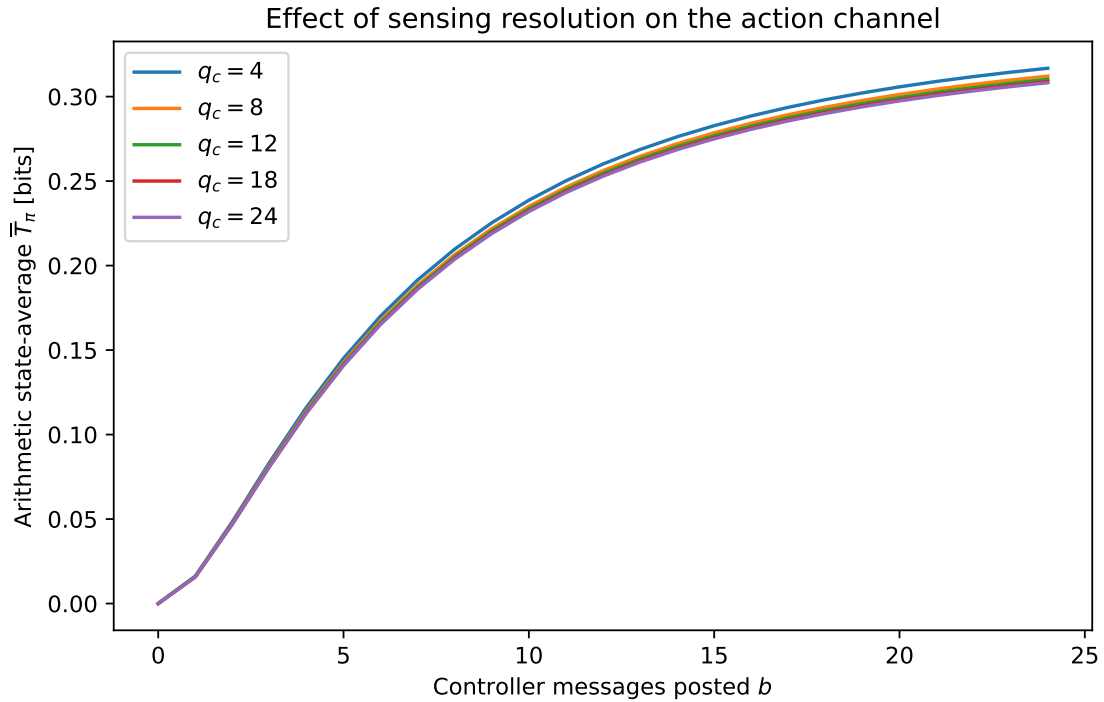


Figure 13: Arithmetic state-average $\bar{T}_\pi$ while varying the controller sensor size q_c . Sensing changes where the action remains variable across states; it does not directly change the blackboard response kernels.

A second useful phase plane is the microscopic response plane (p_1, γ_1) . The same T_π can arise from very different mean responses because information detects any distributional separation, not only motion of the mean.

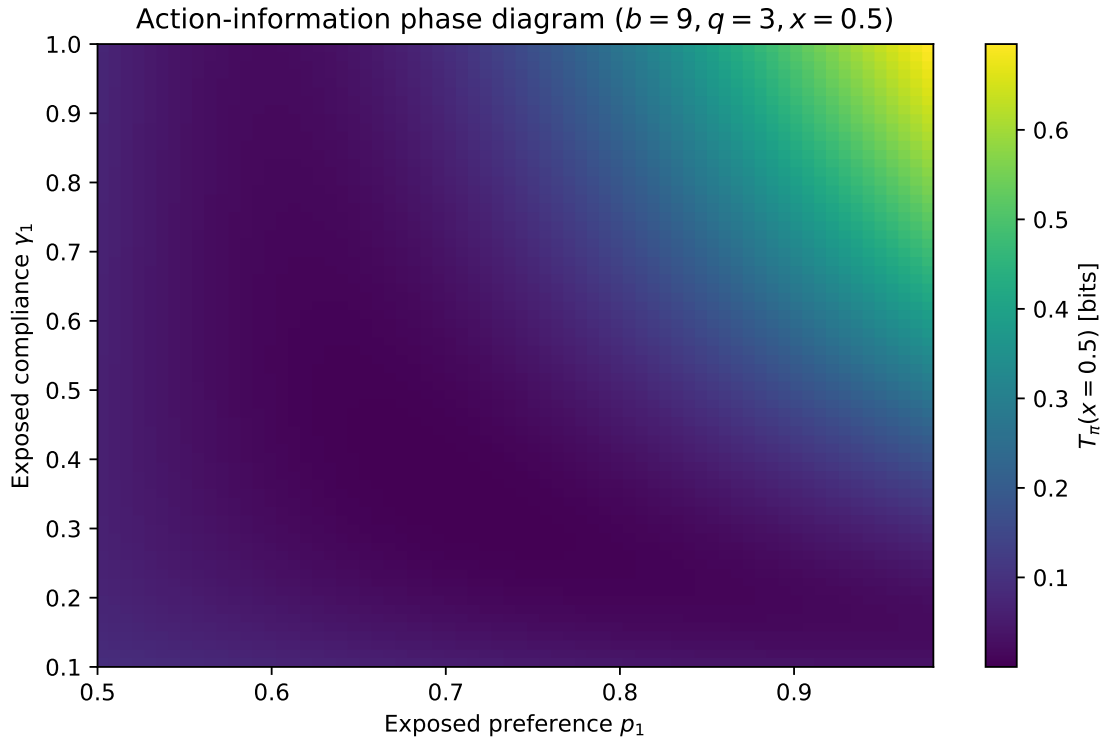


Figure 14: Action-information phase diagram at $x = 0.5$. The action channel can be informative even near the zero-susceptibility contour because intervention and silence may alter fluctuations or higher moments without producing a large mean displacement.

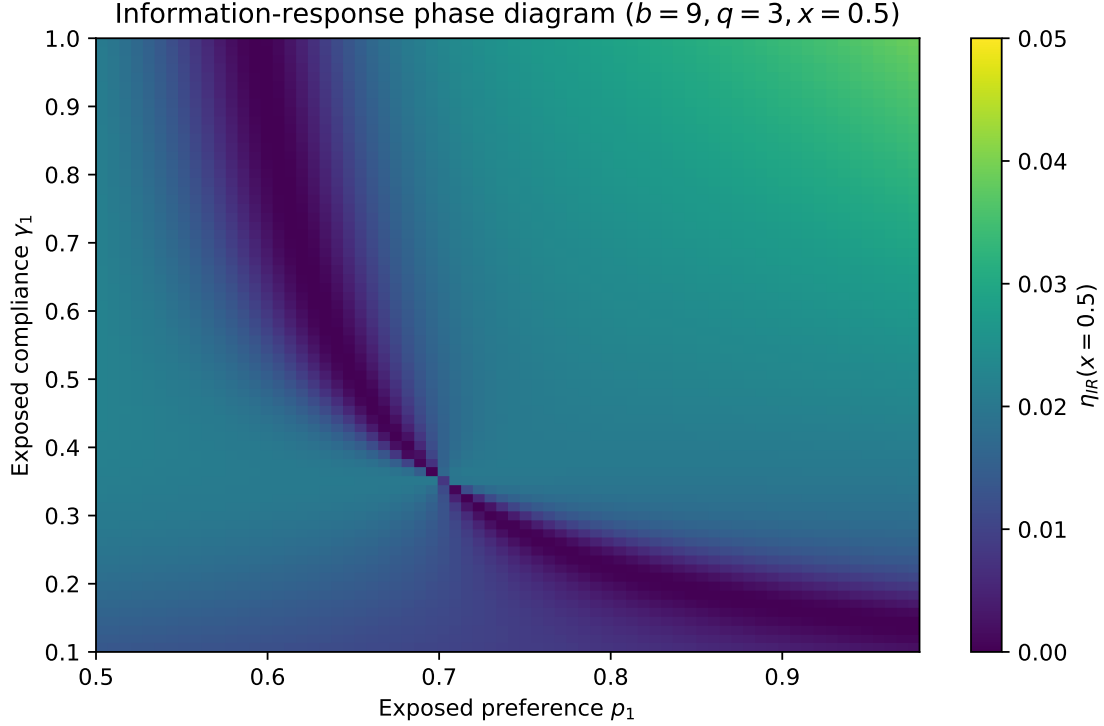


Figure 15: Information-response efficiency in the same (p_1, γ_1) plane. Comparing this plot with the previous one separates “the action changes the distribution” from “the change is coherently directional.”

6.5 How to read the parameter dependence

The blackboard model separates three bottlenecks:

1. **decision bottleneck:** the sensor and policy determine whether the controller posts, through a_n ;
2. **exposure bottleneck:** B, b, q determine whether a focal update actually sees controller content, through w_b ;
3. **response bottleneck:** $(\gamma_0, p_0, \gamma_1, p_1)$ determine how much exposed and unexposed updates differ.

Large T_π requires all three to cooperate. A large message budget is ineffective if q is too small or the board too crowded. High exposure is ineffective if $K_1 \approx K_0$. Strongly different kernels can still yield low T_π when the feedback action is nearly deterministic and $H(U | n)$ collapses.

7 Controlled current

The earlier isolated-layer definition $U(n' - n)$ was automatically controller-specific because the silence branch was the identity. That is no longer true: under the blackboard model, silence has its

own drift. We therefore define the *controller-excess current* by comparing the two branches at the same initial state.

Let

$$\mu_u(n) = \mathbb{E}[n' - n \mid n, U = u], \quad u \in \{0, 1\}.$$

Then

$$\mu_1(n) - \mu_0(n) = N\chi_b(n/N).$$

For pre-round occupancy $p_k(n)$,

$$J_c^{\text{ex}} = \sum_n p_k(n) a_n [\mu_1(n) - \mu_0(n)] = N \sum_n p_k(n) a_n \chi_b(n/N).$$

This is the expected target-count motion attributable to selecting the active branch instead of the silent branch, under the reference model.

For comparison, the actual mean population current is

$$J_{\text{tot}} = \sum_n p_k(n) [(1 - a_n)\mu_0(n) + a_n\mu_1(n)].$$

The all-silent counterfactual current is

$$J_{\text{sil}} = \sum_n p_k(n) \mu_0(n),$$

and indeed

$$J_{\text{tot}} - J_{\text{sil}} = J_c^{\text{ex}}.$$

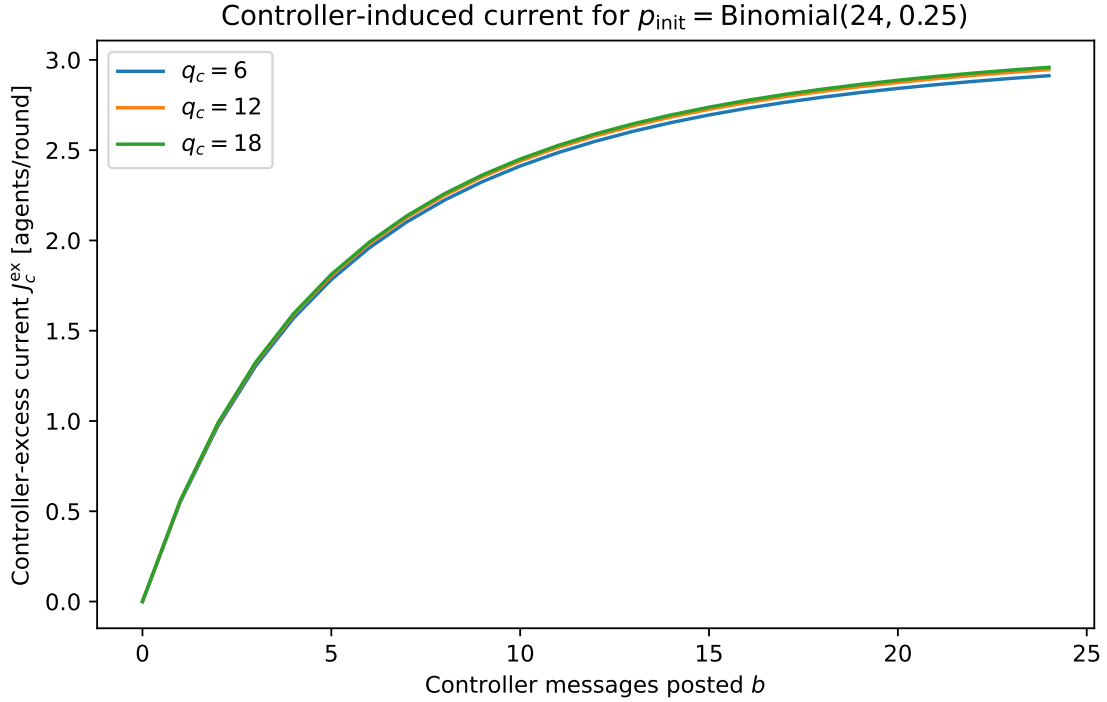


Figure 16: Controller-excess current for the illustrative finite-time ensemble. The increase with message budget is sublinear because exposure saturates. Differences among sensor sizes arise through the state-dependent advocacy probability.

8 Transition-ratio identities for both branches

For any $0 < p < 1$ define the branch affinity

$$h(p) = \ln \frac{p}{1-p}.$$

The finite-compliance factor γ again cancels from the forward/reverse ratio:

$$\frac{K_{\gamma,p}(n+1 | n)}{K_{\gamma,p}(n | n+1)} = \frac{N-n}{n+1} e^{h(p)}.$$

With

$$S_{\text{mix}}(n) = \ln \binom{N}{n},$$

this becomes

$$\ln \frac{K_{\gamma,p}(n+1 | n)}{K_{\gamma,p}(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h(p).$$

Weighted reversibility is preserved by matrix powers. Therefore the round kernels satisfy

$$\ln \frac{Q_u(m | n)}{Q_u(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h_u(m-n),$$

where

$$h_0 = \ln \frac{p_0}{1-p_0}, \quad h_b = \ln \frac{p_b}{1-p_b},$$

and $h_u = h_0$ for $u = 0$, $h_u = h_b$ for $u = 1$.

This is the mathematical reason a finite-time path identity still exists. The important difference is that there are now *two branch affinities*, not a single affinity multiplying a current that vanishes under silence.

9 Forward and reverse finite-time feedback paths

For one feedback cycle, define the forward coarse path measure

$$P_F(n, y, u, m) = p_k(n) S(y | n) \pi(u | y) Q_u(m | n).$$

The final ensemble is

$$p_{k+1}(m) = \sum_{n,y,u} P_F(n, y, u, m).$$

The sensor marginal is

$$p_{Y,k}(y) = \sum_n p_k(n) S(y | n).$$

Use the same reverse-reference construction as in the original report:

$$P_R(m, y, u, n) = p_{k+1}(m) p_{Y,k}(y) \pi(u | y) Q_u(n | m).$$

The reverse experiment begins from the forward final ensemble, samples the stored sensor label from its forward marginal, applies the same policy law, and uses the reverse of the same branch kernel. No stationary distribution is assumed.

10 Finite-time path irreversibility and second-law identity

Define the stochastic log-ratio

$$\Sigma^* = \ln \frac{P_F(n, y, u, m)}{P_R(m, y, u, n)}.$$

The policy factor cancels, yielding

$$\Sigma^* = \ln \frac{p_k(n)}{p_{k+1}(m)} + \ln \frac{S(y | n)}{p_{Y,k}(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}.$$

Define the stochastic coarse system entropy

$$s_{\text{sys}}(n; p) = -\ln p(n) + S_{\text{mix}}(n),$$

and the pointwise sensing information

$$i(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)}.$$

Using the branch transition ratio gives the exact pathwise identity

$$\boxed{\Sigma^* = \Delta s_{\text{sys}} + i(n; y) + h_u(m - n).}$$

Averaging over forward paths,

$$\boxed{\Sigma = \Delta S_{\text{sys}} + I(n_k; Y_k) + \mathcal{A}_k = D_{\text{KL}}(P_F \| P_R) \geq 0,}$$

where

$$\boxed{\mathcal{A}_k \equiv \mathbb{E}_F[h_U(n_{k+1} - n_k)] = \sum_n p_k(n) [(1 - a_n)h_0\mu_0(n) + a_nh_b\mu_1(n)].}$$

The system-entropy functional is exactly the same coarse combinatorial quantity used before:

$$S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}.$$

For the illustrative baseline with $b = 9$ and $p_{\text{init}}(n) = \text{Binomial}(24, 0.25)$, the direct path KL and the decomposition agree to numerical precision: the maximum susceptibility error against explicit matrix powers is 1.3×10^{-15} and the path-identity residual is about 2.3×10^{-14} nats.

11 What thermodynamic efficiency survives?

This section is the most important conceptual difference from the earlier report.

11.1 An exact total path-transport fraction

Separating the transient storage term gives

$$C_{\text{path},k} \equiv \Sigma_k - \Delta S_{\text{sys},k} = \mathcal{A}_k + I(n_k; Y_k).$$

Whenever $\mathcal{A}_k \geq 0$ and $C_{\text{path},k} > 0$, the ratio

$$\eta_{\text{path},k} = \frac{\mathcal{A}_k}{\mathcal{A}_k + I(n_k; Y_k)} \leq 1$$

is mathematically bounded.

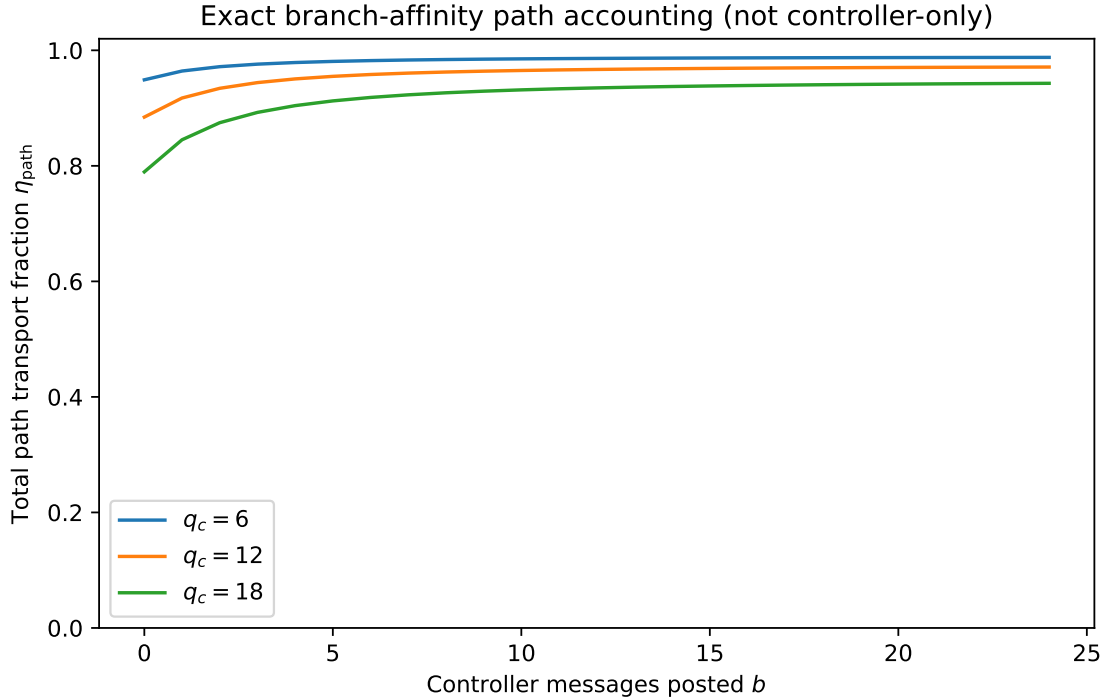


Figure 17: Exact total path-transport fraction for the illustrative baseline. It has the same algebraic form as the earlier thermodynamic efficiency, but its numerator includes baseline social transport and therefore it is not a controller-only efficiency.

11.2 Why this is not the old controller thermodynamic efficiency

In the isolated-actuation report, $Q_0 = I$. Therefore silence generated no population current and no affinity-weighted transport. The non-storage term could be identified cleanly as

$$hJ_c + I(n; Y),$$

with hJ_c entirely attributable to controlled actuation.

Here, $Q_0 = K_0^M$ is generally nontrivial. The affinity-flow term can be decomposed as

$$\mathcal{A}_k = h_0 J_{\text{tot},k} + (h_b - h_0) J_{\text{adv},k},$$

where

$$J_{\text{adv},k} = \sum_n p_k(n) a_n \mu_1(n).$$

The first term contains baseline social transport. Relative to the all-silent counterfactual,

$$\mathcal{A}_k - h_0 J_{\text{sil},k} = h_0 J_{c,k}^{\text{ex}} + (h_b - h_0) J_{\text{adv},k}.$$

This is an exact algebraic excess-affinity-flow diagnostic, but it is not guaranteed to be nonnegative and it does not by itself define a second-law-bounded controller expenditure. Therefore the original quantity

$$\eta_{\text{th}} = \frac{h J_c}{h J_c + I(n; Y)}$$

should not be reused unchanged for the general blackboard model. Doing so would silently assign autonomous baseline dynamics to the controller.

This is not a failure of the blackboard theory. It is a useful result: once both branches evolve, thermodynamic attribution becomes a separate modeling problem rather than an automatic consequence of the path identity.

11.3 A sufficient special case: neutral baseline affinity

There is one clean case in which the path fraction regains a controller-associated interpretation. If the baseline response has

$$p_0 = \frac{1}{2} \quad \Longleftrightarrow \quad h_0 = 0,$$

then the silent branch contributes no directional affinity term. The exact path identity becomes

$$\Sigma = \Delta S_{\text{sys}} + I(n; Y) + h_b J_{\text{adv}},$$

and one may define

$$\boxed{\eta_{\text{th}}^{(h_0=0)} = \frac{h_b J_{\text{adv}}}{h_b J_{\text{adv}} + I(n; Y)} \leq 1}$$

for target-directed operation. This is a *sufficient modeling condition*, not a property of the general blackboard model.

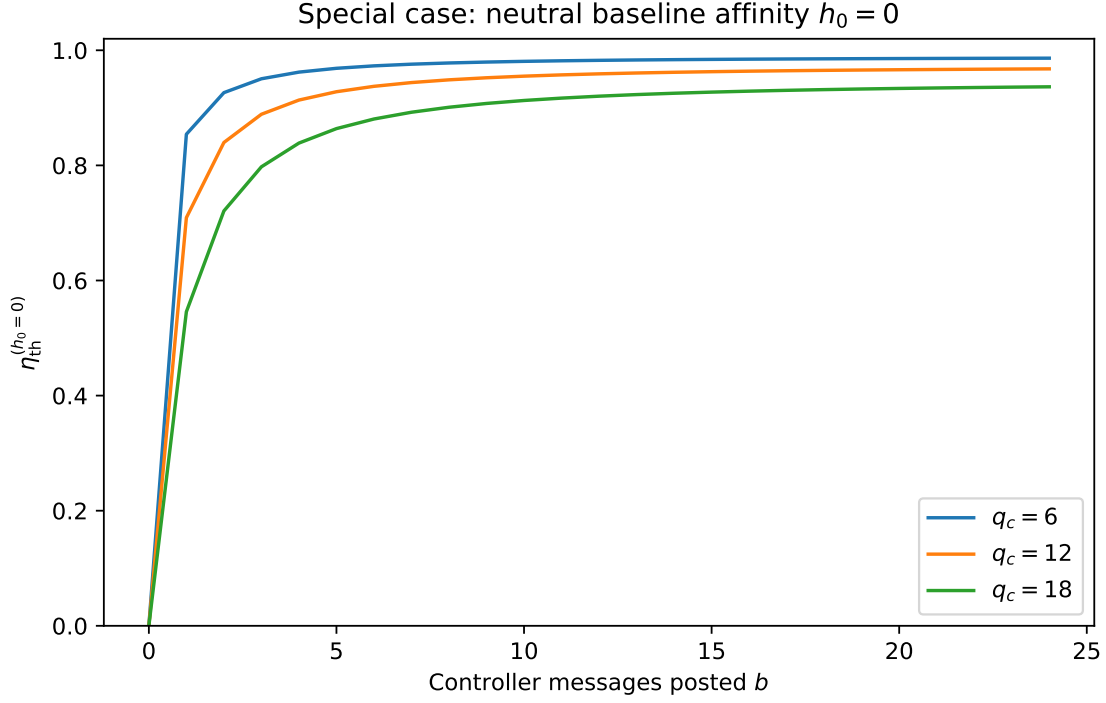


Figure 18: Thermodynamic efficiency in the special neutral-baseline case $h_0 = 0$. Only under such an additional attribution assumption is the bounded path ratio naturally controller-associated.

11.4 Current–irreversibility view as a complementary diagnostic

Even without a unique controller-only thermodynamic efficiency, the pair

$$(J_c^{\text{ex}}, \Sigma)$$

is well defined and useful. It asks how much controller-induced response is obtained while the *entire* coarse feedback process incurs a given finite-time irreversibility.

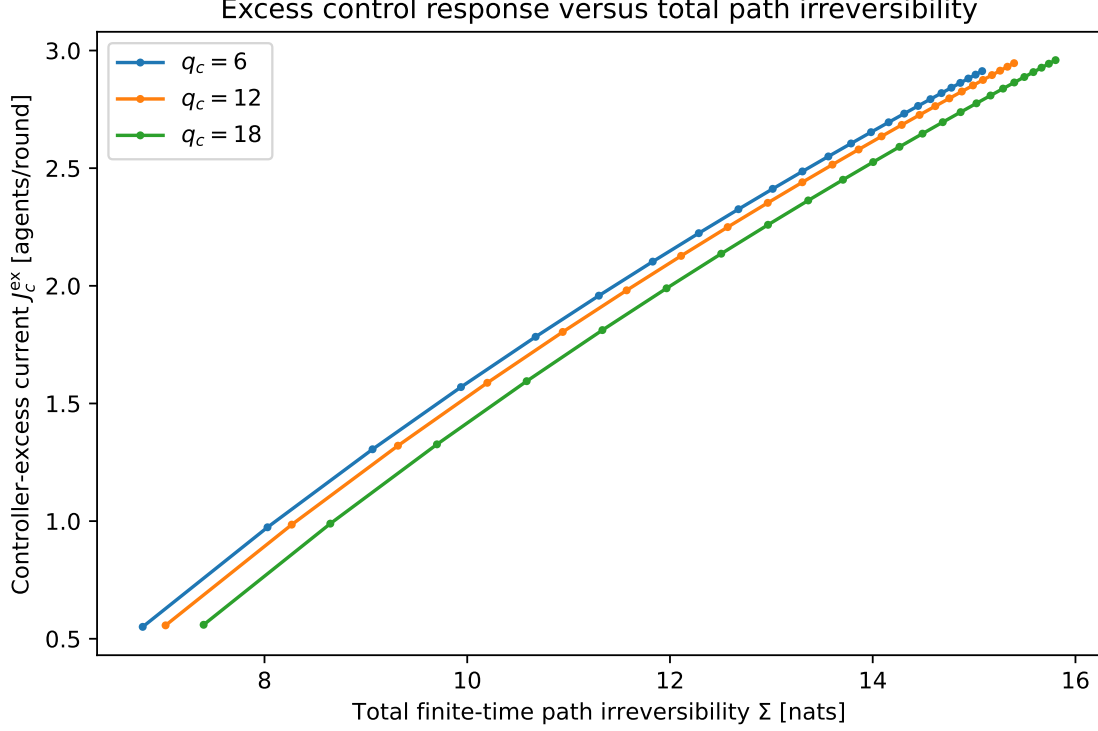


Figure 19: Controller-excess current versus total path irreversibility. Different sensing resources can produce similar excess steering while generating different total irreversibility because the sensing term and baseline population transport differ.

For control design, one can therefore still consider Pareto questions such as

$$\pi^* = \arg \min_{\pi} \Sigma(\pi) \quad \text{subject to} \quad J_c^{\text{ex}}(\pi) \geq J_{\text{req}},$$

or

$$\pi^* = \arg \max_{\pi} J_c^{\text{ex}}(\pi) \quad \text{subject to} \quad \Sigma(\pi) \leq \Sigma_{\text{max}}.$$

The interpretation is simply more careful: Σ is total coarse path irreversibility, not controller expenditure alone.

12 Direct finite-sum computation recipe

For one cycle with pre-sensing ensemble $p_k(n)$, every quantity in this report is computable by finite sums and finite matrix powers.

1. Compute the sensor kernel $S(y | n)$, policy $\pi(1 | y)$, and

$$a_n = \sum_y S(y | n) \pi(1 | y).$$

2. Compute the blackboard exposure probability

$$w_b = 1 - \frac{\binom{B}{m}}{\binom{B+b}{m}}, \quad m = \min(q, B + b).$$

3. Build $K_0 = K_{\gamma_0, p_0}$ and $K_1 = K_{\gamma_1, p_1}$, then

$$\gamma_b = (1 - w_b)\gamma_0 + w_b\gamma_1, \quad p_b = \frac{(1 - w_b)\gamma_0 p_0 + w_b\gamma_1 p_1}{\gamma_b}.$$

4. Form

$$Q_0 = K_0^M, \quad Q_1 = K_{\gamma_b, p_b}^M.$$

5. Compute the exact susceptibility

$$\chi_b(x) = F_M(x; \gamma_b, p_b) - F_M(x; \gamma_0, p_0),$$

and, for $x < 1$, its availability-normalized companion

$$\chi_{b, \text{avail}}(x) = \frac{\chi_b(x)}{1 - x}.$$

Keep $\chi_{b, \text{avail}}$ masked at $x = 1$ and do not interpret it as a bounded efficiency.

6. Compute $T_\pi(n)$ from the weighted Jensen–Shannon divergence, then

$$\eta_{\text{IF}}(n) = \frac{T_\pi(n)}{H_2(a_n)}, \quad \eta_{\text{IR}}(n) = \frac{2a_n(1 - a_n)\chi_b(n/N)^2}{(\ln 2)T_\pi(n)}.$$

7. Compute the controller-excess current

$$J_c^{\text{ex}} = N \sum_n p_k(n) a_n \chi_b(n/N).$$

8. Propagate the actual ensemble with

$$p_{k+1}(m) = \sum_n p_k(n) [(1 - a_n)Q_0(m | n) + a_n Q_1(m | n)].$$

9. Compute $\Delta S_{\text{sys}} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k]$ and the sensing information

$$I(n_k; Y_k) = \sum_{n, y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y, k}(y)}.$$

10. Set

$$h_0 = \ln \frac{p_0}{1 - p_0}, \quad h_b = \ln \frac{p_b}{1 - p_b},$$

calculate

$$\mathcal{A}_k = \sum_n p_k(n) [(1 - a_n)h_0\mu_0(n) + a_n h_b\mu_1(n)],$$

and evaluate

$$\Sigma_k = \Delta S_{\text{sys}, k} + I(n_k; Y_k) + \mathcal{A}_k.$$

As a numerical validation, compute $D_{\text{KL}}(P_F || P_R)$ directly and verify equality.

11. Report $\eta_{\text{path}} = \mathcal{A}/(\mathcal{A} + I)$ only as a *total path-transport fraction*. Report a controller thermodynamic efficiency only if an explicit attribution assumption, such as $h_0 = 0$, has been adopted and justified.

There is no Monte Carlo approximation in this reference calculation. Monte Carlo or direct counting enters only when the same objects are estimated from LLM trajectories.

12.1 How to estimate the terms from repeated blackboard trajectories

The new model makes the required empirical measurements more concrete.

Exposure first. For every microscopic update, record whether at least one controller message was actually sampled. Compare the empirical exposure frequency $\hat{w}_b$ against the combinatorial prediction. If self-exclusion or board growth makes the eligible board size agent- or time-dependent, compare against the corresponding per-update B_{it} rather than a single global B .

Calibrate two response classes. For unexposed updates estimate

$$p_{+,0} = \mathbb{P}(\text{non-}Z \rightarrow Z \mid \text{unexposed}), \quad p_{-,0} = \mathbb{P}(Z \rightarrow \text{non-}Z \mid \text{unexposed}),$$

and for exposed updates analogously $p_{+,1}$ and $p_{-,1}$. The finite-compliance model implies

$$\boxed{\gamma_r = p_{+,r} + p_{-,r}, \quad p_r = \frac{p_{+,r}}{p_{+,r} + p_{-,r}}, \quad r \in \{0, 1\},}$$

whenever the denominator is nonzero. Equivalently,

$$h_r = \ln \frac{p_{+,r}}{p_{-,r}}.$$

This is the direct blackboard analogue of the original single-affinity calibration, now performed separately for the unexposed and exposed channels.

Test the reference out of sample. Fit shared $(\gamma_0, p_0, \gamma_1, p_1)$ on one subset of trajectories and predict held-out raw and availability-normalized susceptibility across message budgets and initial target shares. Do not fit a separate parameter set to every heatmap cell; that would remove the main predictive content of the model.

Use paired branch comparisons whenever possible. The cleanest causal estimate of $\chi_b(x)$ uses intervention and silence continuations from identical complete population states, including the epistemic state and board contents. Observational exposed/unexposed transitions are useful for channel calibration, but they do not alone prove the causal effect of controller exposure.

13 Role of message exposure and kinetic compliance

The blackboard model has two distinct saturation mechanisms.

First, message exposure saturates:

$$w_b \rightarrow 1 \quad \text{as} \quad b \rightarrow \infty.$$

Therefore adding controller messages eventually produces little additional probability of being seen. This is a genuine resource bottleneck absent from the guaranteed-update theory.

Second, conditional on a fixed effective kernel, finite-time relaxation saturates through

$$1 - \left(1 - \frac{\gamma}{N}\right)^M.$$

Even perfect exposure does not create arbitrary motion if the exposed response class has limited compliance.

The two effects should not be conflated. Increasing b changes w_b ; increasing γ_1 changes what happens after exposure. In experimental terms, the first is a communication-access effect and the second is a behavioral-response effect.

A particularly useful diagnostic is the equal-compliance prediction. When $\gamma_0 = \gamma_1$, all x dependence disappears from χ_b . Thus the observed shape of the susceptibility phase diagram can tell us whether the main blackboard effect is directional preference, changed revision kinetics, or unmodeled hidden state.

14 Operational calibration from microscopic LLM transitions

A practical calibration protocol is:

1. Freeze one task, one prompt family, one board-lifetime rule, one message-sampling rule, and one epistemic regime.
2. For every update record the focal vote before and after, whether a controller message was sampled, the number of eligible peer/controller messages, the exact sampled message IDs, and the controller action.
3. Estimate $\hat{w}_b$ directly and compare it with the combinatorial prediction.
4. Estimate (γ_0, p_0) from unexposed transitions and (γ_1, p_1) from exposed transitions.
5. Use those shared parameters to predict Q_0 , Q_1 , $\chi_b(x)$, $\chi_{b,\text{avail}}(x)$, $T_\pi(n)$, $\eta_{\text{IF}}(n)$, and $\eta_{\text{IR}}(n)$ across held-out budgets.
6. Compare state-resolved theory against data before pooling. Systematic residual structure in x , κ , ϕ , or message type is evidence for a necessary extension rather than a reason to let $p_b(x)$ vary freely.

The blackboard model is especially valuable here because it separates three empirical questions that were previously mixed together: *was the controller active?* *was its message actually seen?* *did seeing it change the revision law?*

15 Scope and modeling limitations

The theory is a finite-state coarse-grained reference for the target-support count. It is not a full thermodynamics of language-model reasoning.

Frozen-board approximation. The exact power $Q_1 = K_b^M$ assumes the same B , w_b , and response parameters throughout the round. Live board growth, message expiration, self-exclusion, and within-round changes in the set of eligible posts violate this assumption. If the per-update exposure probabilities are known, the natural next extension is a product of time-dependent kernels rather than a single matrix power.

Binary exposure. The model records only whether at least one controller message was sampled. It ignores the number, identity, redundancy, and semantic content of the sampled controller posts.

If response depends on those quantities, exposure should be refined into multiple classes rather than hidden in a freely varying $p_b(x)$.

Homogeneous response. The same $(\gamma_0, p_0, \gamma_1, p_1)$ is used for all agents and states. Epistemic state, accumulated facts, private history, or agent identity can violate this. The strongest extension is therefore to condition the response classes on a small number of measured epistemic variables, not to fit an arbitrary state-dependent kernel.

Message budget versus actual posts. The theory uses the number b actually posted. If the controller is allowed to post at most b messages but sometimes posts fewer, the realized post count must be used in w_b .

Thermodynamic attribution. The branchwise finite-time path identity is exact for the reference kernels. What is not automatic is the attribution of all non-storage irreversibility to the controller. Because the silent branch evolves, a general blackboard controller efficiency requires an additional decomposition or a more detailed exposure-resolved path model. The safe bounded efficiencies that require no such attribution are η_{IF} and η_{IR} .

Meaning of the observables. The sensing information $I(n; Y)$ asks how much the controller learns about the current population. The exposure probability w_b asks whether posted control content is likely to be seen. The susceptibility χ_b asks how much the active branch changes the mean relative to silence. Its availability-normalized companion $\chi_{b,\text{avail}}$ expresses that same response relative to the remaining non-target share and is deliberately kept separate from the bounded efficiency measures. The local information T_π asks how visible the controller action is in the next population distribution. The information fraction η_{IF} asks how much of the available action uncertainty survives into the next state. The information-response efficiency η_{IR} asks how much of that distributional change appears as coherent mean motion. The excess current J_c^{ex} aggregates the causal branch contrast over occupied states. Finally, Σ quantifies total finite-time path irreversibility of the coarse feedback process. These quantities answer different questions and should remain separate.

Summary of what changed relative to the isolated-actuation report

The blackboard generalization changes surprisingly little algebraically, but it changes the interpretation substantially:

1. The observable susceptibility remains intervention minus silence.
2. Message budget enters first through an exposure probability w_b ; it is no longer identified with a number of controlled update opportunities.
3. Silence evolves: $Q_0 = K_0^M$ rather than I .
4. Intervention is an exposure mixture: $Q_1 = K_{\gamma_b, p_b}^M$.
5. The susceptibility stays closed form and becomes affine in x .
6. An availability-normalized companion $\chi_{b,\text{avail}} = \chi_b/(1 - x)$ exposes the remaining-population effect; it is a diagnostic and may exceed one.
7. T_π , η_{IF} , and η_{IR} remain exact and bounded.

8. The controller-excess current must subtract the silent branch.
9. A finite-time KL identity still exists, but it contains action-dependent branch affinities.
10. The resulting total path fraction is not automatically a controller thermodynamic efficiency because baseline social dynamics contribute to the same path irreversibility.
11. A controller-associated thermodynamic efficiency is recovered under additional attribution conditions, such as a neutral baseline affinity $h_0 = 0$.

This is the main theoretical payoff of the generalization: the phase diagrams for susceptibility and information remain tractable, while the thermodynamic section tells us exactly which parts of the previous interpretation survive and which require new assumptions.

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